Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/16/2014
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	Constrained Vapor Bubble Flight Experiment II – Binary Mixtures		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2013	**End Date:**	09/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wayner, Peter (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNX13AQ78G		
Performance Goal No.:			
Performance Goal Text:			

The thermophysical principles underlying change-of-phase heat transfer systems controlled by interfacial phenomena under microgravity conditions are not well understood. As a result, related passive engineering systems like heat pipes have not been optimized and it is unclear whether recent advances in the design of these systems based on ground-based research using fluid mixtures can be translated to microgravity systems. Herein, we propose basic experimental and theoretical studies of the nonisothermal constrained vapor bubble (CVB-2) using a mixture of pentane and isohexane under microgravity conditions to help remedy this undesirable situation. This proposed study is multi-faceted: 1) it is a basic scientific study in interfacial phenomena, microgravity fluid physics and thermodynamics; 2) it is a basic study in thermal and mass transport; and 3) it is a study of a passive heat exchanger. Although the basic scientific Facets (1) and (2) are emphasized, the research naturally leads to the development of a passive heat exchanger optimized for microgravity conditions.
The proposed optical study of a vapor bubble constrained in a transparent glass cell under nonisothermal conditions will

Task Description:	increase the basic understanding of transport systems controlled by interfacial phenomena. The pressure gradient field will be obtained optically whereas the temperature field will be obtained using thermal sensors. Due to the sensitivity of systems of the proposed relatively large size to gravity, to small temperature and pressure gradients, and to changes in fluid composition, these transport systems need to be studied under microgravity conditions. An "axisymmetric" system with a small Bond number is required. Direct extensions to the microgravity environment first investigated in CVB-1 are proposed. The augmented Young-Laplace equation and the Kelvin-Clapeyron model will be used to help understand and quantify how the system operates. The immediate macroscopic objectives of CVB-2 are to determine the stability, the fluid flow characteristics, the average heat transfer coefficient in the evaporator and condenser sections, and the heat conductance of the CVB as a function of the heat input using the new fluid mixture. The immediate microscopic objective is to determine the detailed characteristics of the transport processes in the curved liquid film. The local conditions under which cavitation and instability occur with the formation of a dry region will be determined as a function of heat flux, film thickness, and stress. High fluxes and relatively large systems with small capillary pressures will be emphasized. The data using the pentane/isohexane mixture will be compared with the pure pentane data obtained in CVB-1 and ground-based data on identical fluid mixtures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work addresses fundamental questions regarding phase-change heat and mass transfer operations that are becoming increasingly important in a wide variety of fields ranging from energy conversion, to microelectronics cooling, MEMs devices, self-assembly, and space flight. Of particular concern to NASA are microchannel and heat pipe systems used for thermal management and perhaps separation processes. Such systems are attractive for long-term space flights because they generally require no moving parts and are potentially maintenance free. The ability to control transport processes in an extended meniscus will be of great utility both on Earth and in space, whether we are looking for enhanced heat transfer, coating, or self-assembly applications. Some of our preliminary ground-based experiments concerning the microscopic evaporating meniscus, performed as part of the existing CVB-1 flight experiment program, have indicated that using a fluid mixture, with the minority component being the less volatile, higher surface tension liquid, offers the opportunity to control fluid flow and heat transfer in an evaporating meniscus and thus enhance the performance of heat pipes. The surface concentration gradient will also enhance axial fluid flow in the corners. The extension of the CVB-1 experiment to mixtures allows us to isolate the Marangoni effects from the effects of gravity and begin to understand how to exploit Marangoni flows for improving the performance of heat pipes in space and on Earth.
Task Progress:	During the past year of the research, we have been vigorously working on analyzing the data from the CVB1 flight experiments performed during 2010, writing papers, and presenting the information at as many conferences as we could reasonably attend. We have also conducted all the CVB2 flight experiments and are now analyzing that data for presentation and publication. The highlights are listed below and the reference details are listed in the Bibliography section. 1. We have spent much of the time this year regressing and analyzing the data from the high heat input experiments run aboard the ISS in late 2010 using the 30 and 40 mm module experiments on the ISS and developing enhanced thermal models to determine the internal heat transfer rates within the cuvette. This has resulted in one submitted publication to the International Journal of Heat and Mass Transfer. 2. We have analyzed the high heat input experiments and have shown that the decrease in performance of the heat pipe is not related to dryout or boiling limitations but instead results from flooding of the hot end. This has been a result never before seen in heat pipes and results from Marangoni forces at the hot end. We are working on a publication to be submitted by the end of summer 2014. 3. We have run all the flight experiments associated with CVB2 and have observed much different behavior than with CVB1. Experiments stopped when the CVB controller module failed. The module will be returned to Earth during the Fall of 2014. The preliminary results and observations from the CVB2 experiments will be analyzed during the next fiscal year. 4. Abstracts for 5 meeting presentations were approved.
Bibliography Type:	Description: (Last Updated: 10/14/2025)
Abstracts for Journals and Proceedings	Wayner PC Jr, Kundan A, Plawsky JL. "CVB: The Constrained Vapor Bubble Capillary 40 mm Fin Experiment on the ISS" 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts, p. 47-8. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Kundan A, Wayner PC Jr, Plawsky JL. "Explosive Nucleation in Microgravity: The CVB Experiment." American Physical Society Annual Meeting, Denver, CO, March 3-7, 2014. American Physical Society Annual Meeting, Denver, CO, March 3-7, 2014. , Mar-2014
Abstracts for Journals and Proceedings	Kundan A, Wayner PC Jr, Plawsky JL. "Constrained Vapor Bubble (CVB) Capillary Fin Experiments on the ISS." ASME 2014 Fluids Engineering Summer Meeting (FEDSM2014), Chicago, IL, August 3-7, 2014. ASME 2014 Fluids Engineering Summer Meeting (FEDSM2014), Chicago, IL, August 3-7, 2014. , Aug-2014

Abstracts for Journals and Proceedings	Wayner PC Jr, Kundan A, Plawsky JL. "Fluid Flow in the Marangoni Dominated Region of the Constrained Vapor Bubble Heat Pipe." 30th Meeting of the American Society for Gravitational and Space Research (ASGSR), Pasadena, California, October 22-26, 2014. 30th Meeting of the American Society for Gravitational and Space Research (ASGSR), Pasadena, California, October 22-26, 2014. In press, as of July 2014. , Jul-2014
Abstracts for Journals and Proceedings	Kundan A, Plawsky JL, Wayner PC Jr. "The Marangoni Limit to the Operation of Wickless Heat Pipes." 2014 Annual AIChE Meeting, Atlanta, Georgia, November 16-21, 2014. 2014 Annual AIChE Meeting, Atlanta, Georgia, November 16-21, 2014. In press as of July 2014. , Jul-2014
Articles in Peer-reviewed Journals	Plawsky JL, Fedorov AG, Garimella SV, Ma HB, Maroo SC, Chen L, Nam Y. "Nano- and microstructures for thin-film evaporation—a review." Nanoscale and Microscale Thermophysical Engineering, 2014. In press as of July 2014. , Jul-2014
Journal/Magazine covers	Plawsky JL, Fedorov AG, Garimella SV, Ma HB, Maroo SC, Chen L, Nam Y. "Cover of Special Issue on Nano and Microscale Phase Change Phenomena in journal Nano and Microscale Thermophysical Engineering." Nanoscale and Microscale Thermophysical Engineering, 2014. In press as of July 2014. , Jul-2014